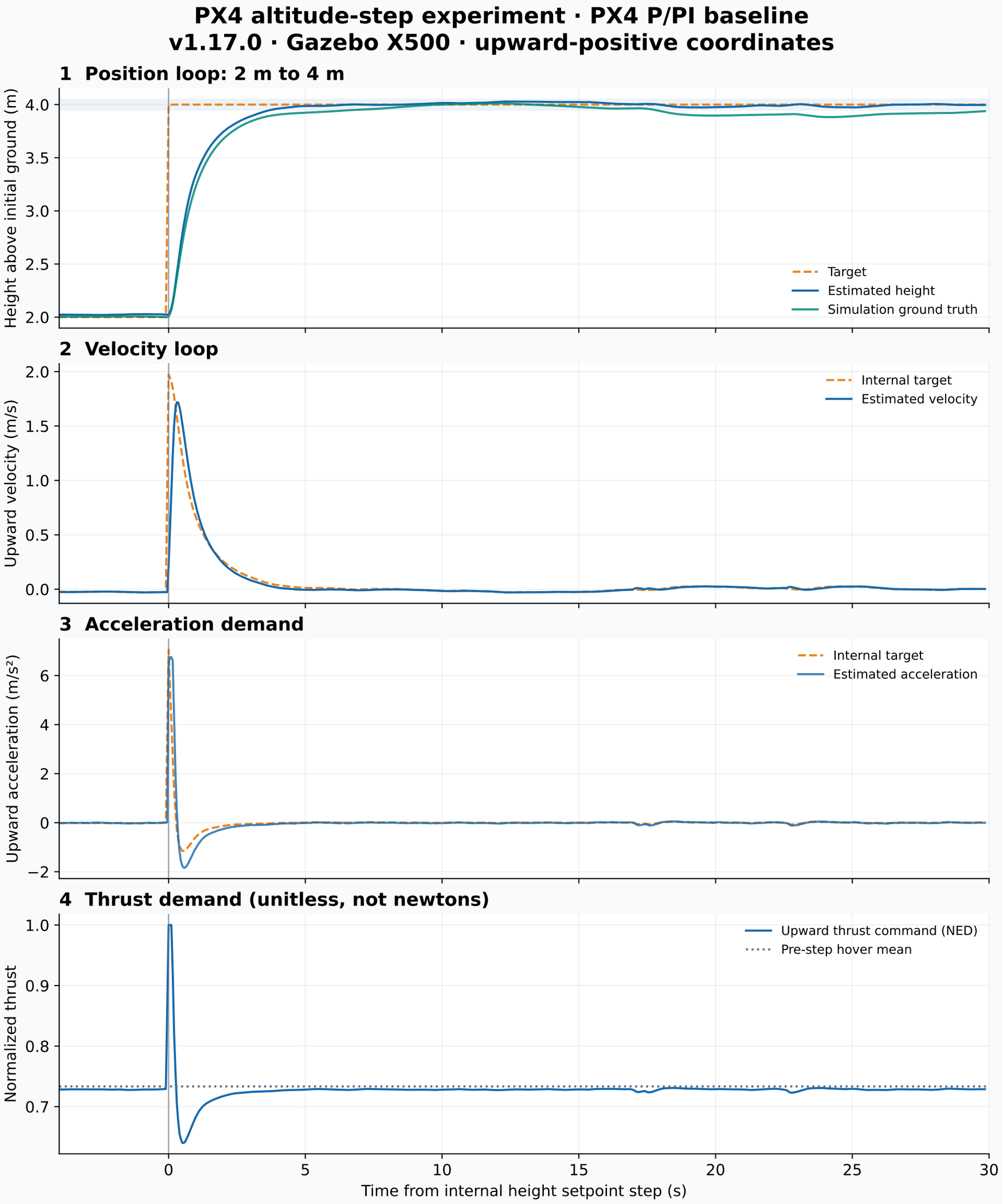
PX4 altitude-step experiment · PX4 P/PI baseline
v1.17.0 · Gazebo X500 · upward-positive coordinates
1 Position loop: 2 m to 4 m
Height above initial ground (m)
4.0
3.5
3.0
2.5
2.0
Target
Estimated height
Simulation ground truth
2 Velocity loop
Upward velocity (m/s)
2.0
1.5
1.0
0.5
0.0
Internal target
Estimated velocity
3 Acceleration demand
Upward acceleration (m/s²)
6
4
2
0
−2
Internal target
Estimated acceleration
4 Thrust demand (unitless, not newtons)
Normalized thrust
1.0
0.9
0.8
0.7
Upward thrust command (NED)
Pre-step hover mean
0
5
10
15
20
25
30
Time from internal height setpoint step (s)